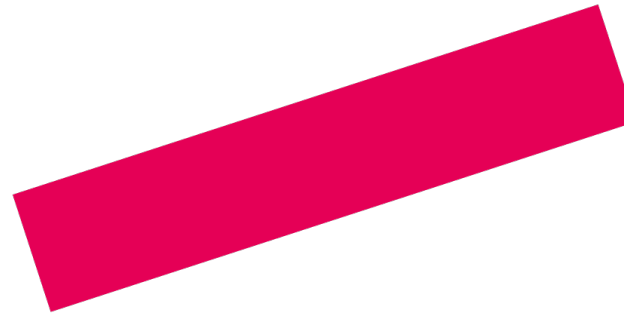


_OUTLIERS



LECTURE OVERVIEW

- **Lesson goals**
- **Outliers**
- **Workshop**
- **Missing Data**

LESSON GOALS

- The student identifies outliers and differentiates between outliers and erroneous data (relates to qualifications: data understanding, data analytics).
- The student identifies missing data and interprets its meaning (relates to qualifications: data understanding, data analytics).
- The student reflects on what outliers mean in their data (relates to qualifications: data understanding, data analytics).
- The student distinguishes between "dirty data" and choosing the wrong distribution model (relates to qualifications: data understanding, data analytics).

OUTLIERS (1): WHAT IS AN OUTLIER ANYWAY?

- What is an outlier anyway?
- Example 1.
- What is an outlier is related to the shape of the distribution.
- Matplotlib boxplots flag values as outliers as follows:
 1. Calculate the InterQuartile Range (IQR): $Q3 - Q2$
 2. $> Q1 - 1.5 \times \text{IQR}$ and $< Q3 + 1.5 \times \text{IQR}$ are outliers (ie the lowest 25% and the highest 25%)
- Alternate method: calculate the Z-score:
 - $z = \frac{(x - \mu)}{\sigma}$
 - $|z| > 3$: outliers
- What constitutes an outlier is a matter of convention.

OUTLIERS (2): ASSUMPTIONS

- If you flag the lowest 25% and the highest 25% as outliers, there should be a clearly recognizable middle 50%.
- The Z-score also makes assumptions about the shape of the data.
- Other assumptions:
 - Data comes from a single population
 - Data is continuous
 - Observations are independent
- What could happen if one or more of these assumptions is not met?

OUTLIERS (3): IF DATA IS NOT NORMALLY DISTRIBUTED

- Try a different method for calculation outliers
 - EG: Mean Absolute Deviation:
 1. Calculate the absolute deviation of each data point from the **median** and then apply a scaling factor to it.
 2. Outliers are all points with $MAD_{scaled} > 2.5$ (or some other value)
- Turn the distribution into a normal distribution
 - EG: Apply a log transformation to a log normal distribution and **then** calculate the IQR.
- Adjust the cutoff points for the IQR
 - Useful if the distribution is skewed
- Create subgroups and apply outlier detection within each subgroup
 - Do this if the data is multi modal.

OUTLIERS (3): OUTLIERS ARE NOT ERRONEOUS DATA

- **Even if the data is normally distributed, the fact that a data point is an outlier does not in itself mean anything!**

Outliers	Erroneous data
Unusual but valid observations	Invalid data points
Can provide valuable insights	Should be corrected or removed
Domain specific	Validation rules
May indicate important phenomena	Measurement / entry errors

OUTLIERS (4): BEST PRACTICES

- Investigate before removing:
 - Check for data entry / coding errors.
 - Verify measurement accuracy.
 - Make sure you understand the domain.
- Document decisions:
 - Record which values were modified and why.
 - Be transparent about criteria used.
 - Make sure your work is reproducible
- Sensitivity analysis:
 - Run analyses with and without outliers.
 - Report differences in results.

OUTLIERS (extra): TIME SERIES

- Time series can have outliers, too.
- BUT: make sure your time series is stationary before you start investigating them (why?).

WORKSHOP

1. Detecting outliers and deciding what to do with them.
2. Turning non-normal distributions into normal distributions

MISSING DATA (1): MCAR, MAR and MNAR

- MCAR: Missing Completely At Random
- MAR: Missing At Random
- MNAR: Missing Not At Random

MISSING DATA (2): MCAR

- MCAR: Missing Completely At Random
 - Missingness is independent of any variables.
 - Least problematic: unbiased results even with deletion.
 - Causes: machine failure, spilled coffee, etc.
 - Example: during patient appointments, doctors sometimes forget to record all the required data.

MISSING DATA (3): MAR

- MAR: Missing At Random
 - Missingness depends on some variable that is observed.
 - Can be dealt with.
 - Causes: you know them.
 - **Still random!**
 - Example: patients miss appointments if they are elderly, but you know their age so you can correct for this.

MISSING DATA (4): MNAR

- MNAR: Missing Not At Random
 - Missingness depends on unobserved variables.
 - Most problematic: will require further work that often involves domain knowledge.
 - Causes: these are not known, which is why this is so problematic.
 - Example: patients miss appointments but it is unknown whether this simply because they are old (and you don't know their ages to begin with).

MISSING DATA (5): EXAMPLE

- Example: Titanic data set

MISSING DATA (6): HOW TO DEAL WITH MISSING DATA

- DropNA
- Imputation: filling missing data with sensible values
 - Simple:
 - Mean / Median / Mode substitution
 - Last observation carried forward
 - Not so simple:
 - Simple prediction (regression)
 - k-Nearest Neighbors
 - Maximum Likelihood Estimation
 - Random Forest Imputation
 - Etc
 - See <https://scikit-learn.org/stable/modules/impute.html>

MISSING DATA (7): IMPUTATION RISKS

- Imputation risks:
 - Reduces variance
 - May introduce bias
 - Distorts relationships between variables
 - Obscures the fact that missingness is itself data

MISSING DATA (8): IMPUTATION EXAMPLE

- Imputing missing age values in the Titanic data set.

MISSING DATA (9): IMPUTATION BEST PRACTICES

- Always understand WHY data is missing.
- Choose methods appropriate to the situation:
- Missing Completely At Random: simple methods may suffice
- Missing Not At Random: collect additional data or use sensitivity analysis
- Compare the effect of different imputation methods